

SightSeeker

Tell it what to find. It sees, decides and drives there.

Problem Statement 01 · Vision-Based Autonomous Robot

YOLO26	30 / 30	Wokwi
real detection on my webcam	missions in the 3D rover twin	rover firmware simulated

TEAM

SYNTAX TERROR (COS)

Harshit · solo builder

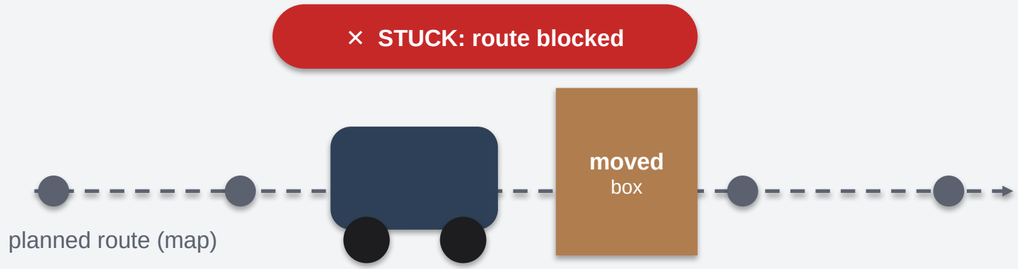
- Vision AI: YOLO26 + goals
- Autonomy and steering
- Firmware and circuit
- Simulation and 3D twin

PROBLEM STATEMENT & THEME

PS 01 · VISION-BASED AUTONOMOUS ROBOT

Most mobile robots follow a fixed map. When the room changes they get stuck, and only an engineer can give them a new task.

What happens today



Maps go stale

one moved box breaks the route

Hard to re-task

new job = engineer reprograms

Costly sensing

LiDAR robots cost lakhs

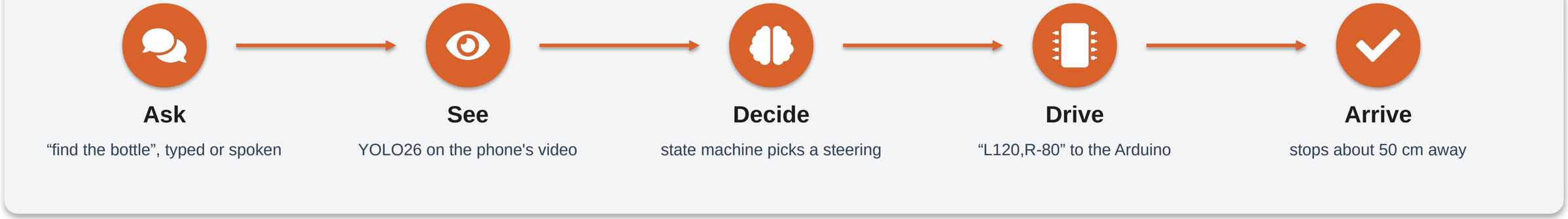
Impact Small warehouses, labs and hospitals can't afford a mapping project for every change. A camera rover that understands its goal works the same day.

Theme alignment: what PS 01 asks for, and where it is in SightSeeker

✓ **Object detection** YOLO26	✓ **Scene understanding** objects + distance	✓ **Language goals** “find the bottle”
✓ **Obstacle avoidance** camera + HC-SR04	✓ **Motor control** L298N wheel PWM	✓ **Real-time decisions** state machine
✓ **Autonomous navigation** search → reach	✓ **Sensors** camera, ultrasonic	✓ **Safety** watchdog, 25 cm stop

SOLUTION

Say what to find. SightSeeker sees it, decides how to get there and drives, with safety checks on the rover itself.



Unique value: compared with today's options

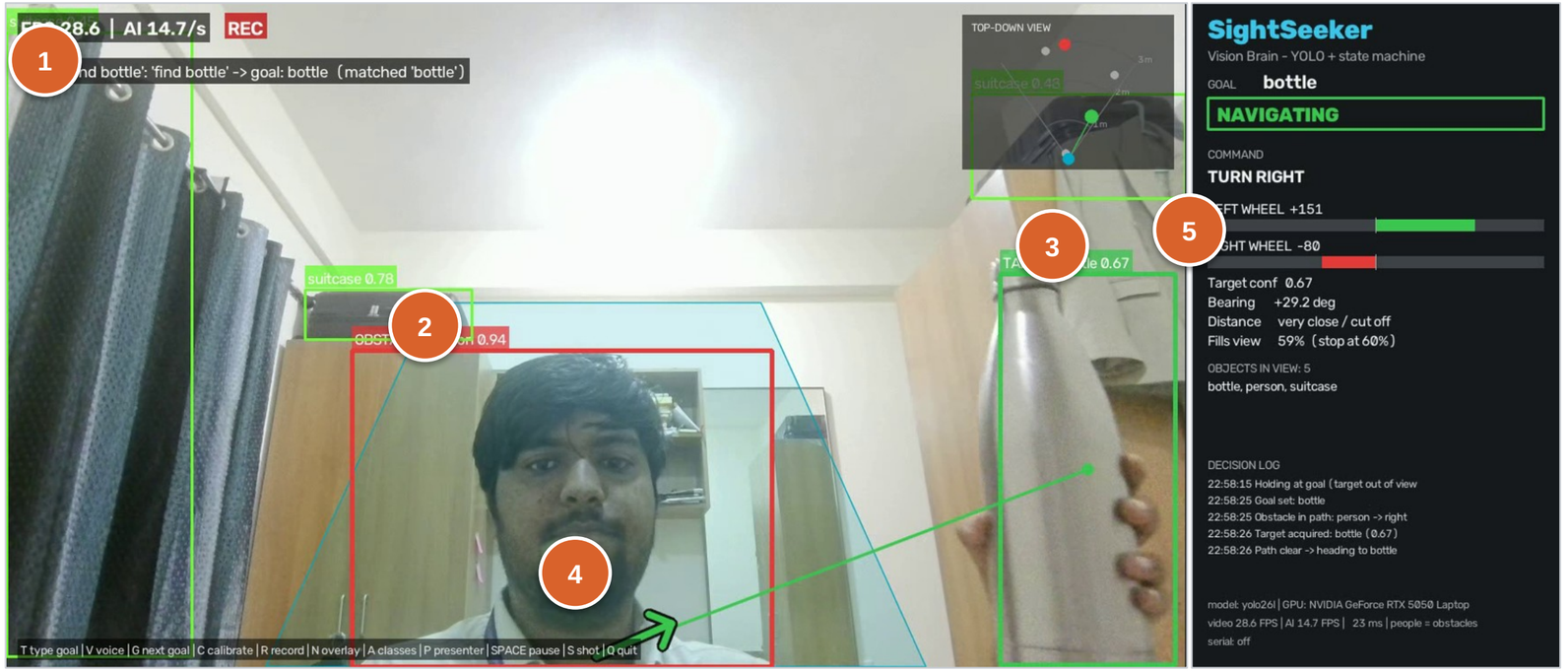
	Line follower	Map-based robot	SightSeeker
Setup	tape on the floor	mapping run	**none: just a camera**
New task	re-lay the tape	engineer reprograms	**say it in English**
Moved objects	✕ stops	re-map needed	**✓ sees and avoids**
Cost	low	lakhs	**≈ ₹1,500 + laptop**

Key features

- Plain-English and voice goals
- Walk mode: guides a person by voice
- Digital twin and 3D rover twin
- 3 safety layers on the Arduino

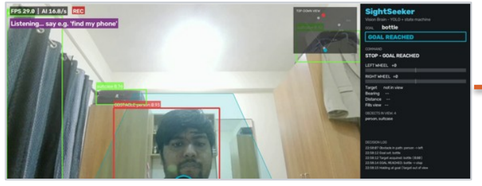
PROTOTYPE: HOW IT WORKS

A real run on my laptop: YOLO26 on the GPU, 28 frames per second, webcam as the robot's eye

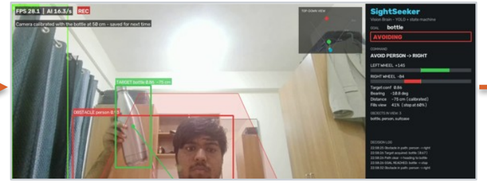


1. **Hears the goal**
   "find bottle" → class bottle
2. **Person in the path**
   marked as an obstacle
3. **Target locked**
   bottle, 67% confidence
4. **Steering line**
   heading to the bottle
5. **Wheel commands**
   left +151, right −80: turn right

The same run, start to finish



1 Waiting for a goal

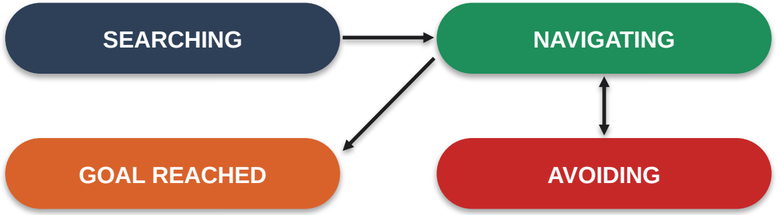


2 Avoiding the person



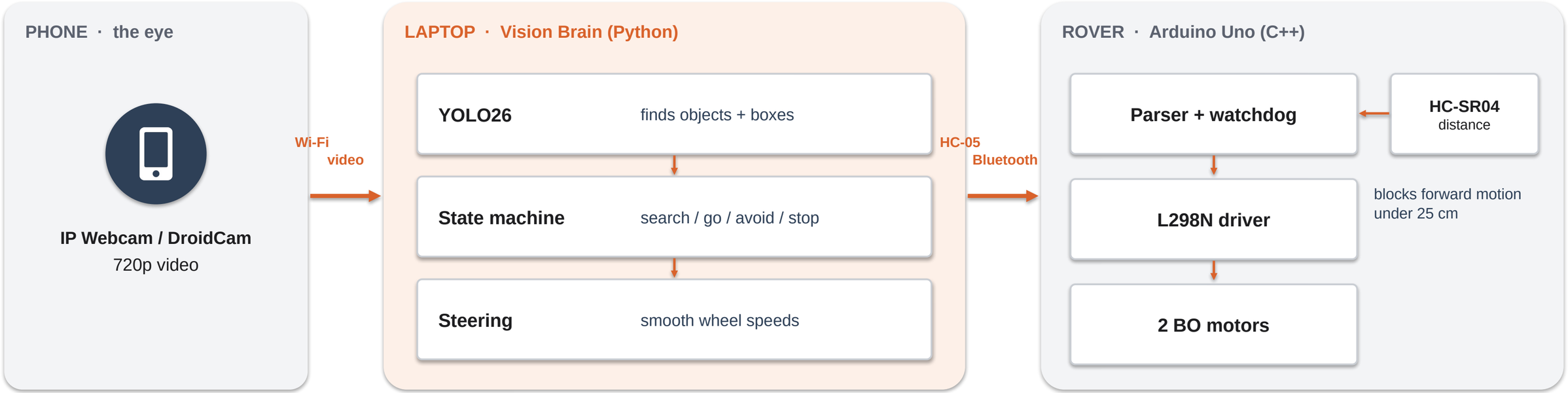
3 Goal reached in 8.6 s, 51 cm away

Decision loop (state machine)



ARCHITECTURE

The phone sees, the laptop thinks, the rover acts. One short text line connects them.



Serial line, 20× a second: L120,R-80 (left, right: −255…255)

Tech stack

Python · OpenCV · Ultralytics YOLO26 · PyTorch + CUDA · Arduino C++ · Three.js · Wokwi

Safety runs on the rover, not the laptop

- No command for 500 ms → motors stop
- Obstacle under 25 cm → no forward
- Every value clamped to ±255

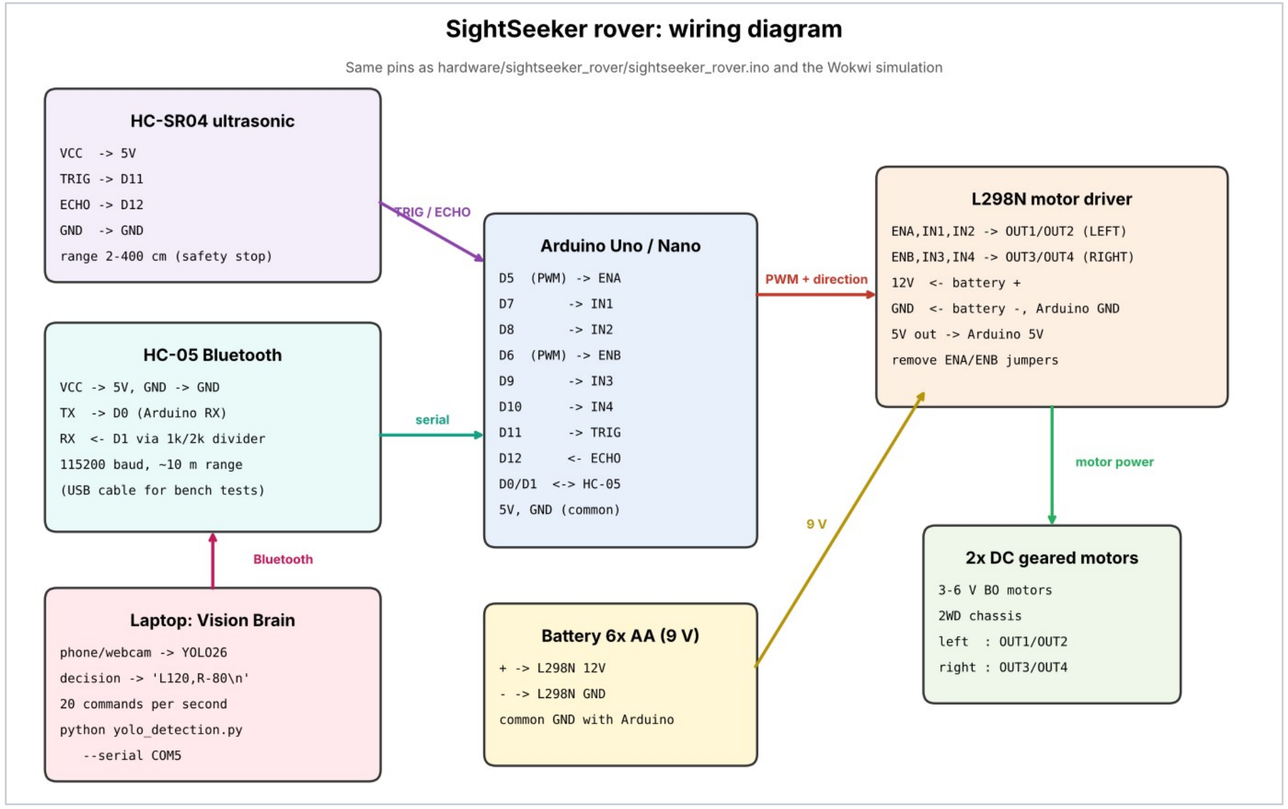
APIs and data

No cloud and no paid APIs: everything runs offline on the laptop.

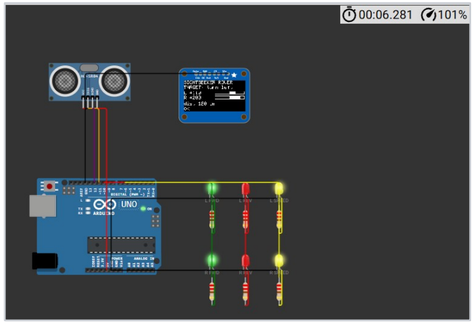
The same commands drive the digital twin and walk mode.

HARDWARE: CIRCUIT & SIMULATION

Arduino Uno + L298N + HC-SR04 + HC-05, wired on paper and running in Wokwi

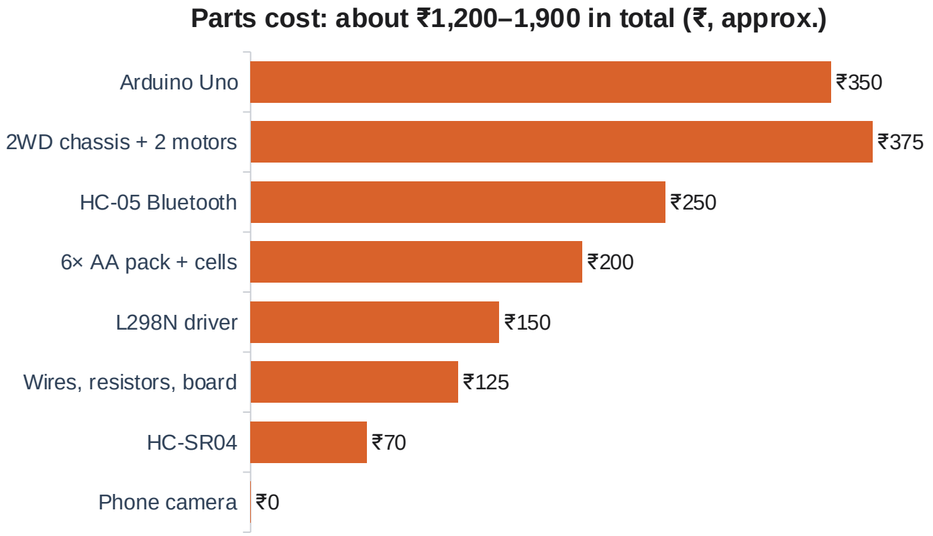


Wiring diagram: the same pins are used in the firmware, Wokwi and the 3D twin



My Wokwi run: Uno, HC-SR04, OLED and LEDs on the L298N pins

```
Wokwi serial monitor
[demo] SEARCH: scan right
[demo] TARGET: turn left
[demo] NAVIGATE: forward
[demo] LINK LOST: no cmds
[safety] no command for
  500 ms -> motors stopped
```



CURRENT PROGRESS: WHAT WORKS TODAY

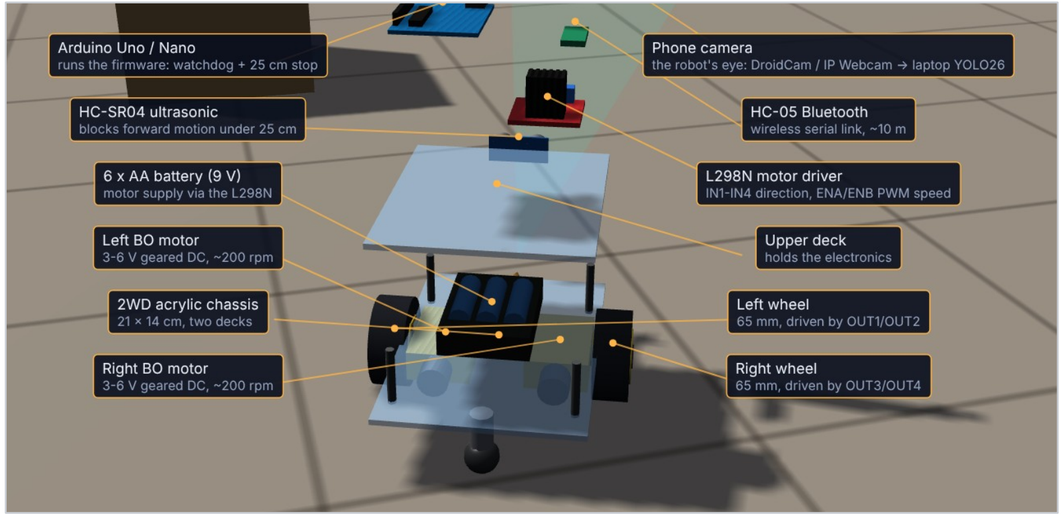
Software, firmware and simulations are built and tested. The physical rover is the next step.

- ✓ Vision Brain
- ✓ Rover firmware
- ✓ Wokwi simulation
- ✓ 3D rover twin
- ✓ Walk mode + twin
- → Physical rover

30 / 30	3D twin missions reached the bottle
8.6 s	real webcam run: goal to target, 3 obstacles avoided
1,000 / 1,000	earlier warehouse simulation, 0 collisions



3D rover twin, mode 1: phone view, serial line and L298N pins

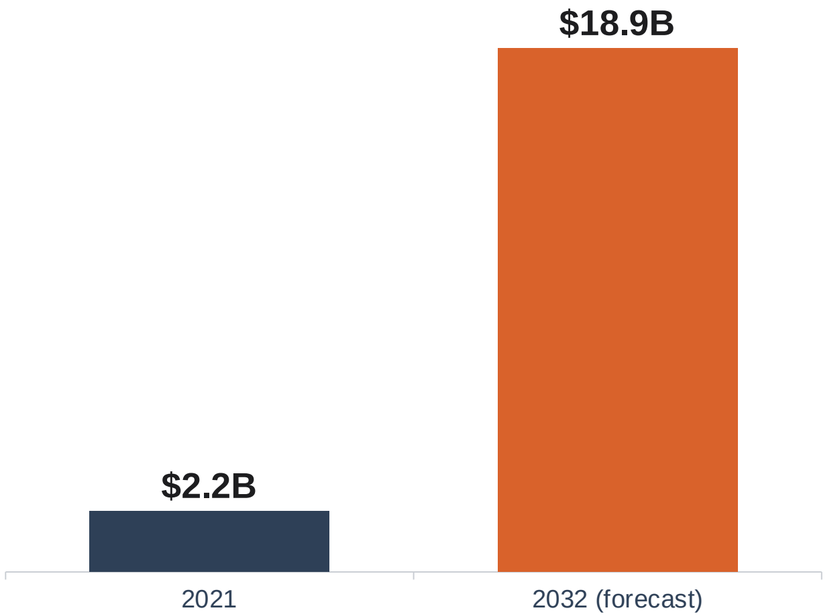


Exploded view: every part, with its job and pins

MARKET RESEARCH / ANALYSIS

Mobile robots are spreading from big factories to smaller sites, and cost is what holds them back

Global autonomous mobile robot market (US$ billion)



21.8% a year (2022–2032). Source: Allied Market Research. Other firms forecast less, e.g. MarketsandMarkets $7.1B by 2032.

Target users

Colleges and labs
low-cost autonomy kit

Small warehouses
fetch without a mapping job

Hospitals
move supplies around people

Factories
shuttle parts between stations

Visually impaired users
walk mode: voice guidance

Revenue model

1. **Education kits**
   rover + software for colleges (first market)
2. **Software licence**
   vision navigation for existing robots
3. **Custom training**
   detect a client's own objects
4. **Fleet dashboard**
   monthly subscription

FUTURE SCOPE

From simulation to a real rover, then to any object and many robots

NOW · Oct 2026

- YOLO26 Vision Brain
- Firmware in Wokwi
- 3D rover twin, walk mode

NEXT · Build

- Assemble the ₹1,500 rover
- Tune on a real floor
- Record the real demo

THEN · Smarter

- Any object by name (YOLOE, prototyped)
- Phone app for goals
- Custom-trained objects

LATER · Scale

- On-board Jetson / Raspberry Pi
- ROS 2 and warehouse systems
- Fleets sharing one map

The roadmap covers: mobile app · AI model improvement · enterprise integrations · scaling to many users

Web3 India | builder base | EXIDO

hack in hills



Thank you!

Code + docs

github.com/harshittalreja/SightSeeker

Live Wokwi simulation

wokwi.com/projects/479147376881059841